Supporting Systematic Literature Reviews in Computer Science

The Systematic Literature Review Toolkit

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ABSTRACT

The systematic analysis of related work is an important part of every research project. Surprisingly, especially in computer science, this activity is typically performed manually. Whilst in other disciplines fully automated analysis approaches exist, the lack of a reasonably complete, queryable and free-to-use literature catalog for computer science requires computer scientists to manually retrieve, merge and analyze literature from different catalogs. Although manual literature retrieval is enforced by the current main publishers, tool support to merge and cleanse literature found in different catalogs as well as to analyze the results is possible and needed. The systematic literature review toolkit, an Eclipse Rich Client Platform application leveraging the Eclipse Modeling Framework, aims to provide this support. Currently, four main features are supported: simple literature filtering, design of a taxonomy, classification of literature and analysis of the classification by generated diagrams. The tool is available online: https://github.com/sebastiangoetz/slr-toolkit

CCS CONCEPTS

• **Applied computing → Digital libraries and archives**;

KEYWORDS

Systematic Literature Survey, Eclipse, EMF, Xtext

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1 INTRODUCTION

Literature management is an essential part of every research project. It comprises various intermediate actions which can be categorized into three main activities: information gathering, knowledge extraction and analysis. [1] Information gathering denotes the identification of relevant literature catalogs, to search them for relevant literature and to filter the search results (e.g., by in-/exclusion criteria) to retrieve an initial literature corpus. Knowledge extraction denotes systematic reading of the literature and its metadata to extract the essence relevant for the current research project.

Finally, analysis denotes reasoning about the extracted essence, e.g., to compare it with the current research project (demarcation) or to identify research questions not, yet, addressed (gap analysis).

While these general steps can be seen as common across all research disciplines, their practical realization differs.

Problem 1: No common catalog. An essential prerequisite for efficient literature management is a common, queryable catalog, where all relevant literature of the respective research discipline is available and correctly annotated with metadata. Without correct metadata, a (semi-)automated analysis is impossible or requires the manual correction of the metadata. Without being queryable by software or providing export functionality, the (semi-)automated processing is rendered impossible, too. Surprisingly, especially for research in computer science, no such catalog exist. Instead, two main approaches are followed. First, literature is retrieved from Google Scholar, which leads to the inclusion of non-peer-reviewed literature and heterogeneous metadata. Second, literature is retrieved from the special catalogs: ACM digital library, IEEE Xplore, Elsevier SCOPUS, Springer Link or the Digital Bibliography & Library Project (DBLP) to name the main sources. Here, typically, only peer-reviewed literature can be found and the quality of metadata is high for each individual catalog. But, the metadata schemes differ between the catalogs. Thus, for both approaches, researchers have to manually check, correct and align the metadata to enable a (semi-)automated processing of the literature corpus.

Problem 2: Tools exist, but most are for other disciplines. As will be pointed out in section 4, a plethora of tools for literature management exists. But, in particular tools for analysis or tools supporting the whole process of literature management are hardly applicable to the computer science context. This might be a direct consequence of the first problem: without a common catalog the effort to retrieve a processable literature corpus is either rendered infeasible or the whole process of literature management is performed manually using standard software. By this, existing tools have no reason to support requirements special to the field of computer science (e.g., support for latex and bibtex).

As the first problem has economic roots and, hence, is unlikely to be solved in the near future, computer scientists need tool support to work with literature from multiple, (intentionally?) different catalogs. While tools for collection and knowledge extraction already exist (e.g., JabRef and Zotero), analysis tools which integrate with them are rare.

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The SLR Toolkit, presented in this paper, denotes such an analysis tool. It allows (1) to import literature corpi in the form of bibtex bibliographies, (2) the specification of a common taxonomy to be used for all corpi, (3) the classification of literature according to this taxonomy and (4) the generation of analysis diagrams based on the classification.

For this, the tool leverages model-driven techniques. The taxonomy is realized as a domain specific language (DSL) using Xtext. The classification of literature is realized by parsing bibtex bibliographies into models adhering to a metamodel specified using the Eclipse Modeling Framework (EMF), which is connected to the metamodel of the taxonomy DSL. From a user's perspective, the tool does not differ from other tools, which are not build using model-driven techniques. But, for developers, it represents a reasonably sized application to test the feasibility of model-driven techniques.

The remainder of this paper is structured as follows. The next section gives an overview of the main features offered by the SLR toolkit. Section 3 outlines the architecture of the tool as well as its build- and release-management. The tool is put into context of the existing tool landscape in Section 4. Finally, Section 5 concludes the paper.

2 FEATURES

The features of the tool can be seen on Youtube[1]. Currently, the tool supports four main features: importing bibliographies, specifying a taxonomy, classifying literature and generation of analysis diagrams.

Figure 1 depicts the user interface of the tool. It shows an exemplary project on models@run.time, which is part of a literature survey currently under review, covering 272 papers. It can be found in the examples folder of the GitHub project of the tool[2]. The perspective shows 6 views.

The left upper view, the "Bibtex Entries" view, enlists all documents part of the current study. By selecting an entry in this list, the bibtex overview in the upper center shows the details for this document and the taxonomy view in the upper right corner shows the current classification for this entry. Using the check boxes of the taxonomy, the classification of the currently selected document can be edited.

Which documents are shown in the "bibtex entries" view is denoted by the bibtex files part of the project shown in the "project explorer" view in the left lower corner. The taxonomy specification is part of the project as .taxonomy file. To edit the taxonomy, either this file can be edited in a text editor or, preferably, the provided refactorings and change operations available as context menu in the taxonomy view can be used. The following operations are supported:

- Creation of new terms, either as siblings or children of existing terms.
- Removal of existing terms.
- Renaming terms.
- Moving a term from one superterm to another. For example, moving the term "requirements" from "level of abstraction" to "model type".
- Splitting a term into multiple subterms. For example, splitting the term "quality" into the terms "performance", "energy" and "usability".

The first two operations (creation and removal) are general change operations, which have a direct impact on the classification and require manual intervention. The last three operations are refactorings and, when executed, the tool automatically updates the classification accordingly. For example, when renaming a term, all documents, which have been classified with this term are adjusted accordingly.

The chart view (lower center), depicts the diagrams of the classification. Currently, three types of diagrams are supported: bar charts, bubblematrix charts and pie charts. Bar charts are automatically shown, whenever the user selects a term with subterms in the taxonomy view. For BubbleMatrix charts exactly two terms with subterms have to be selected. In general, the configuration wizard of the chart view can be used to specify the exact representation of a diagram.

Figure 2 depicts an exemplary settings page for a bubble matrix chart comparing the terms "Model Type" and "Purpose". The "series settings" tab allows to specify, which subterms are to be shown in the diagram and which coloring scheme is to be used. The other two tabs allow to set the title, font, outline as well as the labels on the axes. The tool offers a PDF export (as scalable vector graphic), so the diagrams configured here can be directly used in papers or theses.

As a convenience feature, the tool also supports to import bibtex entries from Google Scholar. After specifying the search query (as in Google Scholars advanced search mask), the import is started as a background process, which will be shown in the right lower view (progress view). As Google Scholar does not want its search results to be automatically processed, this feature might be broken soon.

Finally, the tool supports to synchronize its projects with a Mendeley account. Whether a bibtex file is connected with a Mendeley folder or not, is depicted as gray or green circle overlay on the respective file in the project explorer view.

Features, which are currently under development and are planned to be released in September 2018 are:

- Project Metadata Support (authors, abstract of study, etc.) and Latex Export for Common Templates (ACM, IEEE, Springer)
- Detecting and repairing data quality issues (e.g., broken classification) via the Eclipse quickfix mechanism
- Bibtex Comparison Tooling (merge, diff, overlap)

Features, planned to be released in later releases are:

- Support for Synchronization with https://www.zotero.org
- Improved Visualizations
- CSV Export of Classification Data
- Visualization of Relations between Publications

[1] https://youtu.be/IB4d9CJt144
[2] https://github.com/sebastiangoetz/slr-toolkit

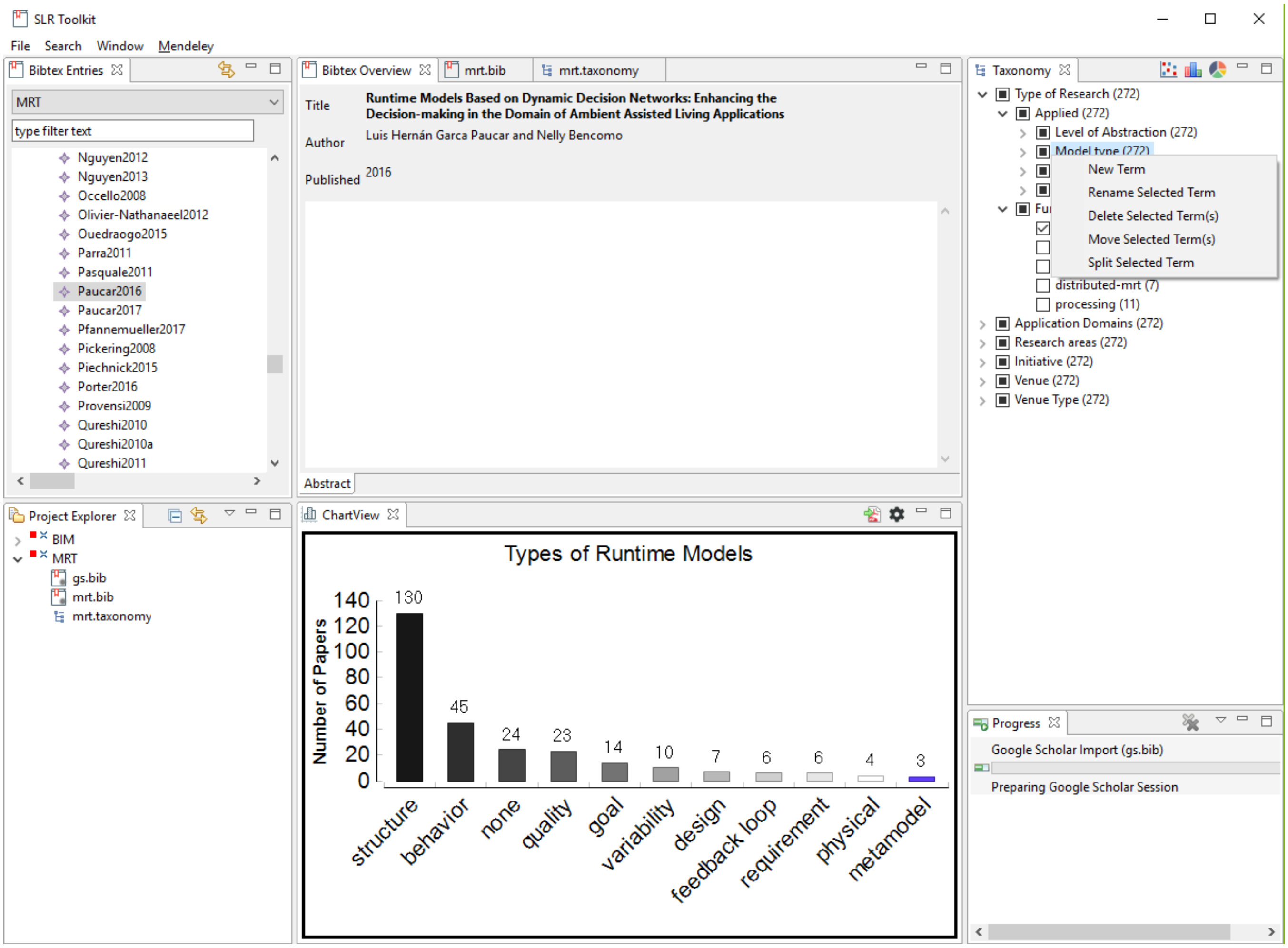

Figure 1: Main Perspective of the Systematic Literature Review Toolkit

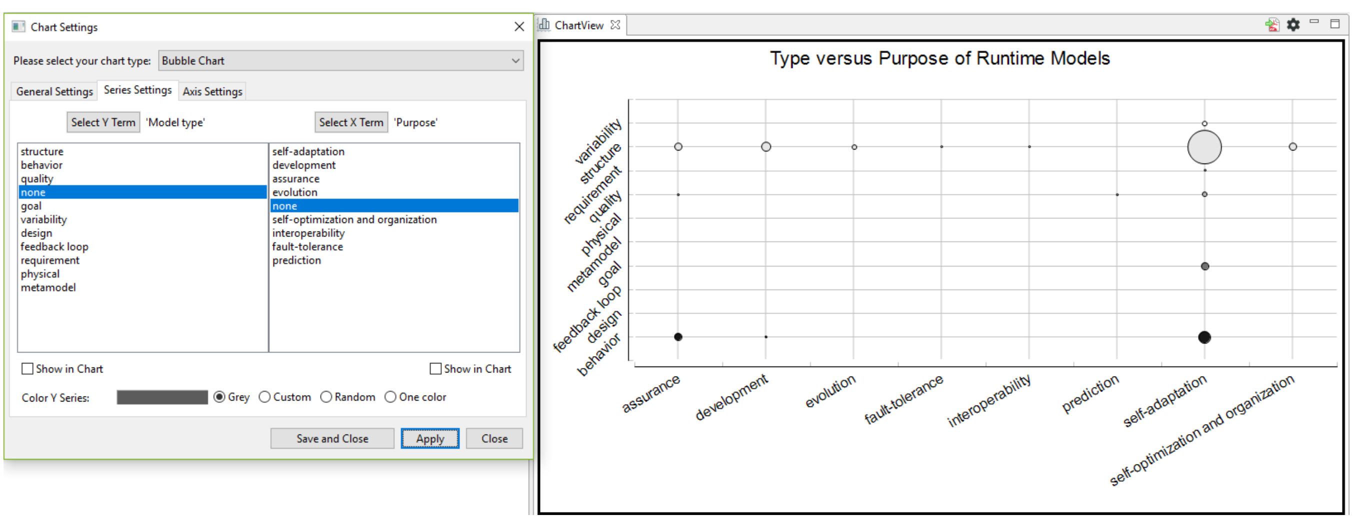

Figure 2: Exemplary Diagram Settings for BubbleMatrix Chart

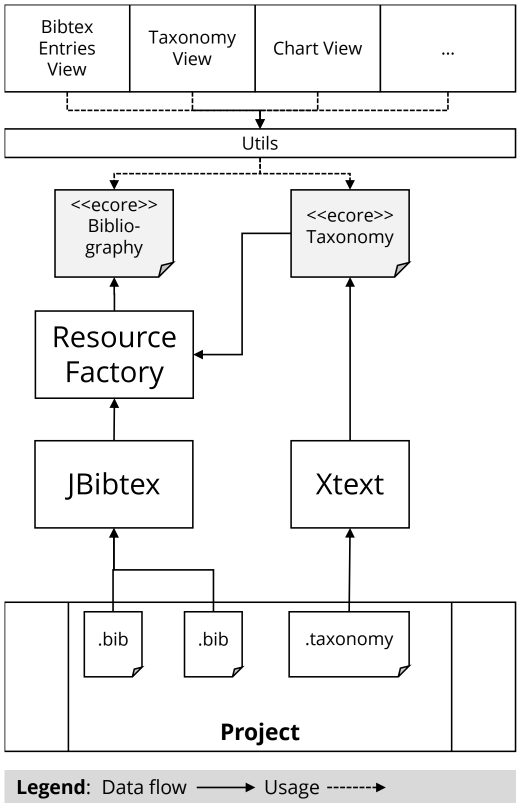

Figure 3: Architecture of the SLR Toolkit

3 ARCHITECTURE, BUILD- AND RELEASE-MANAGEMENT

The SLR toolkit is an Eclipse Rich Client Platform (RCP) application and is comprised of the following main parts:

- Bibtex Processing (`de.tudresden.slr.model.bibtex`)
- Taxonomy Management (`de.tudresden.slr.model.taxonomy`)
- Charting based on Eclipse BIRT (`de.tudresden.slr.ui.chart`)

In the following, first an architectural overview is given, followed by a description of the current build- and release-management.

3.1 Architectural Overview

The overall architecture of the toolkit is depicted in Figure 3. It can be read from top to bottom or vice versa. The data flow from project files to views of the graphical user interface (GUI) is depicted from bottom to top, while the usage starting with the views of the GUI is depicted vice versa.

To avoid the complexity of building a custom bibtex parser (e.g., using Xtext), we use JBibtex[3] to parse and serialize bibtex bibliographies. All entries in all bibliographies of a project are parsed into an Ecore model, which models those parts of the bibliography, which are relevant for the toolkit. This includes a link to another Ecore model, which represents the taxonomy created by a parser generated using Xtext. The resource factory of the bibliography model resolves the links between bibtex entries and their classification.

The classification information is stored in a special field (called "classes") of a bibtex entry. By this, all information of a study remains in one place. Although there is a separate taxonomy specification file, this file can be derived from the classification information in the bibliography file (except for unused classes of the taxonomy). This allows for an easy adoption of versioning systems like Git.

3.2 Build- and Release-management

The SLR toolkit uses Maven as build management tool and leverages Eclipse Tycho for the RCP build. It is hosted as open source software on GitHub under Eclipse Public License. For continuous integration we use Travis CI. By this, every commit to every branch is automatically build and the success or failure is indicated immediately.

The toolkit is developed since summer term 2015 mainly by students as part of their studies. Each student project focuses on a specific feature (e.g., taxonomy refactorings) and works on a separate feature branch. Once the feature has been implemented and the CI build succeeds, the student creates a pull request for the feature to the master branch and, by marking the commit with a version tag, initiates an automated release.

In consequence, depending on the number of students participating per semester, every 6 months (March and September) new releases of the SLR toolkit are available.

4 RELATED WORK

A plethora of tools for literature management and reviews exists. In [7], Marshall and Brereton summarize several approaches, which support researchers in conducting systematic literature reviews. For each approach, the name, the latest publication, a short description, a URL of the prototype (if applicable) and the last update/release of the prototype are given in the following.

- SLR-Tool [5], a tool to support the whole process of conducting a systematic literature review including the classification task
  https://alarcos.esi.uclm.es/slrtool/, 2010.
- Systematic Literature unified Review program (SLuRp) [2], another tool for the whole process
  https://bugcatcher.stca.herts.ac.uk/SLuRp, offline.
- State of the Art through Systematic Review (StArt) [6], another tool for the whole process
  http://lapes.dc.ufscar.br/tools/start_tool, 2013.
- Projection Explorer (PEx) [8], a tool to visualize and explore projections of multi-dimensional data points
  https://github.com/magsilva/sysrevpex, 2014.
- Hierarchical Cluster Explorer [9], a tool to explorer multi-dimensional data
  https://www.cs.umd.edu/hcil/hce/, 2005.
- Site Content Analyzer [3], a tool for text mining, used to automatically extract knowledge from a bibliography
  http://www.sitecontentanalyzer.com/, offline.

[3]https://mvnrepository.com/artifact/org.jbibtex/jbibtex

- UNITEX [12], a tool used to automatically extract result sentences from literature
http://unitexgramlab.org/, 2016.
- ReVis [4], a tool using visual text mining to support the selection task in systematic reviews
http://ccsl.icmc.usp.br/pt-br/projects/revis, offline.
- SLRONT [10], an ontology for systematic literature reviews, no prototype available.
- Linked Data for Selection Process [11], an approach using text mining to semi-automate the selection process of a literature review, no prototype available.

Further tools, which aid researchers in conducting literature reviews are:

- RevMan is a commercial tool (free for academic purposes) supporting the whole process of conducting literature reviews according to the Cochrane Community, a charity organization aiming for better decision making in the healthcare domain
https://community.cochrane.org/help/
tools-and-software/revman-5, 2018.
- Covidence is another commercial tool from Cochrane, which supports researchers in the screening phase of literature reviews
https://www.covidence.org/, 2018.
- Abstrackr is another tool for the screening phase focusing on the healthcare domain provided by center for evidence synthesis in health
https://github.com/bwallace/abstrackr-web, 2018.
- Open Meta-Analyst is another tool from the center for evidence synthesis in health allowing researchers to conduct meta analysis on literature review data
https://github.com/bwallace/OpenMeta-analyst-/, 2016.
- Rayyan is yet another commercial tool supporting the screening phase
https://rayyan.qcri.org/, 2016.
- DistillerSR (commercial) claims to be the world's most advanced systematic literature review software supporting the whole process of a review
https://www.evidencepartners.com/, 2018.

Even more tools can be found using the searchable tool catalog[4] operated by Chris Marshall.

In comparison to this non-exhaustive list of tools and approaches, the tool presented in this paper does only offer a small subset of already available features in other tools. But, the SLR toolkit is the only tool realizing these features with model-driven techniques and, hence, can and shall serve as a testbed for the modeling community.

5 CONCLUSION

In this paper, an Eclipse Rich Client Platform application, which leverages from model-driven techniques (Xtext, EMF), has been introduced. The tool supports researchers in conducting literature reviews by specifying a taxonomy, classifying literature according to it and analyzing the results by generated diagrams.

[4]http://systematicreviewtools.com

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